Yi ZHU

PhD Candidate in Signal and Image Processing | Medical Imaging · Multimodal Learning · 3D Vision
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Profile

PhD candidate in Signal and Image Processing at INSA Lyon, focusing on medical imaging, multimodal learning, and 3D vision. My research includes CBCT-IOS fusion, high-fidelity 3D dental reconstruction, and patient-specific dental digital twins. I have solid experience in PyTorch development, deep learning modeling, and experimental evaluation, with research experience in medical image segmentation, large-scale urban 3D reconstruction, and cross-institutional collaboration. I am able to translate clinical and engineering needs into technical problems, with strong attention to algorithmic accuracy, computational efficiency, and practical impact.

Education

Institut National des Sciences Appliquées de Lyon (INSA Lyon) — **Lyon, France**
PhD Student in Signal and Image Processing — *Oct 2024 – Sep 2027*
Thesis subject: Creation of digital twins by fusion of radiographic and optical data for oral medicine

Institut National des Sciences Appliquées de Lyon (INSA Lyon) — **Lyon, France**
Diplôme d'Ingénieur (Equivalent to a Master's Degree in Eng.) in Electrical Engineering — *Sep 2019 – Aug 2024*

Pohang University of Science and Technology (POSTECH) — **Pohang, South Korea**
Exchange Student in Electrical Engineering with BRMIE Scholarship — *Sep 2022 – Jun 2023*

Research Experience

Fusion of IOS and CBCT — **CREATIS Laboratory, INSA Lyon**
Supervisor: Sébastien Valette, Razmig Kéchichian, Raphaël Richert — *Oct 2024 – Sep 2027*

- Developing multimodal learning methods for high-fidelity 3D dental reconstruction by fusing Cone Beam Computed Tomography (CBCT) and Intraoral Scan (IOS) data.
- Designed a fusion framework that leverages the complementary strengths of CBCT and IOS: CBCT provides root and internal anatomical information, while IOS provides high-resolution crown surface geometry.
- Investigating deep implicit representations and representation learning strategies to reconstruct patient-specific dental digital twins from incomplete and heterogeneous multimodal data.
- Exploring weakly supervised, semi-supervised, and data-efficient learning methods to reduce the dependence on densely annotated dental datasets.
- Evaluating the proposed methods on public dental datasets and hospital-acquired CBCT/IOS data, with the goal of supporting clinical applications in orthodontics, implantology, and endodontics.

3D City Reconstruction Using 360° Panoramic Videos — **National Institute of Informatics, Japan**
Supervisor: Satoshi Ikehata, Kiyoharu Aizawa — *Aug 2025 – Jan 2026*

- Implemented and evaluated SOTA algorithms, including VGGT, MapAnything, and Pi3, to assess their applicability in complex urban scene modeling.
- Analyzed the trade-offs between reconstruction accuracy and computational efficiency, identifying critical optimization paths for deploying these models in real-world large-scale environments.

Evaluation of Algorithms for Medical Image Segmentation — **CREATIS Laboratory, INSA Lyon**
Supervisor: Chantal Muller, Thomas Grenier — *Mar 2024 – Aug 2024*

- Evaluated segmentation models (YOLOv8/v9, UNeXt, DETR/UNETR, U-Net/nnU-Net) using MONAI's Metrics Reloaded (Dice, Hausdorff, ASSD), with focus on lesion-size performance.
- Pretrained on public datasets, then fine-tuned on custom MS data for improved lesion segmentation.

Publications

- ACCV 2026, Poster: MapPano3D: Map-Guided Large-Scale Urban Reconstruction from Panoramic Videos
- MICCAI 2026, STSR Challenge 1st Prize, Oral: Registration Teaches Registration: Transform-Derived Crown Guidance for Semi-Supervised CBCT–IOS Alignment
- ISBI 2026, Poster: High-Fidelity 3D Tooth Reconstruction by Fusing Intraoral Scans and CBCT Data via a Deep Implicit Representation
- ICSP 2024, Oral: Comparative Analysis of Three Advanced Deep Learning Algorithms for Multiple Sclerosis Lesion Segmentation in FLAIR MRI

Skills

- Deep Learning in 3D Vision: PyTorch, VGGT, Pi3, U-Net, Point Transformer, DeepSDF
- Tools: Linux, Git, Docker, CUDA, LaTeX
- Languages: Chinese (Native), French (C1), English (C1)